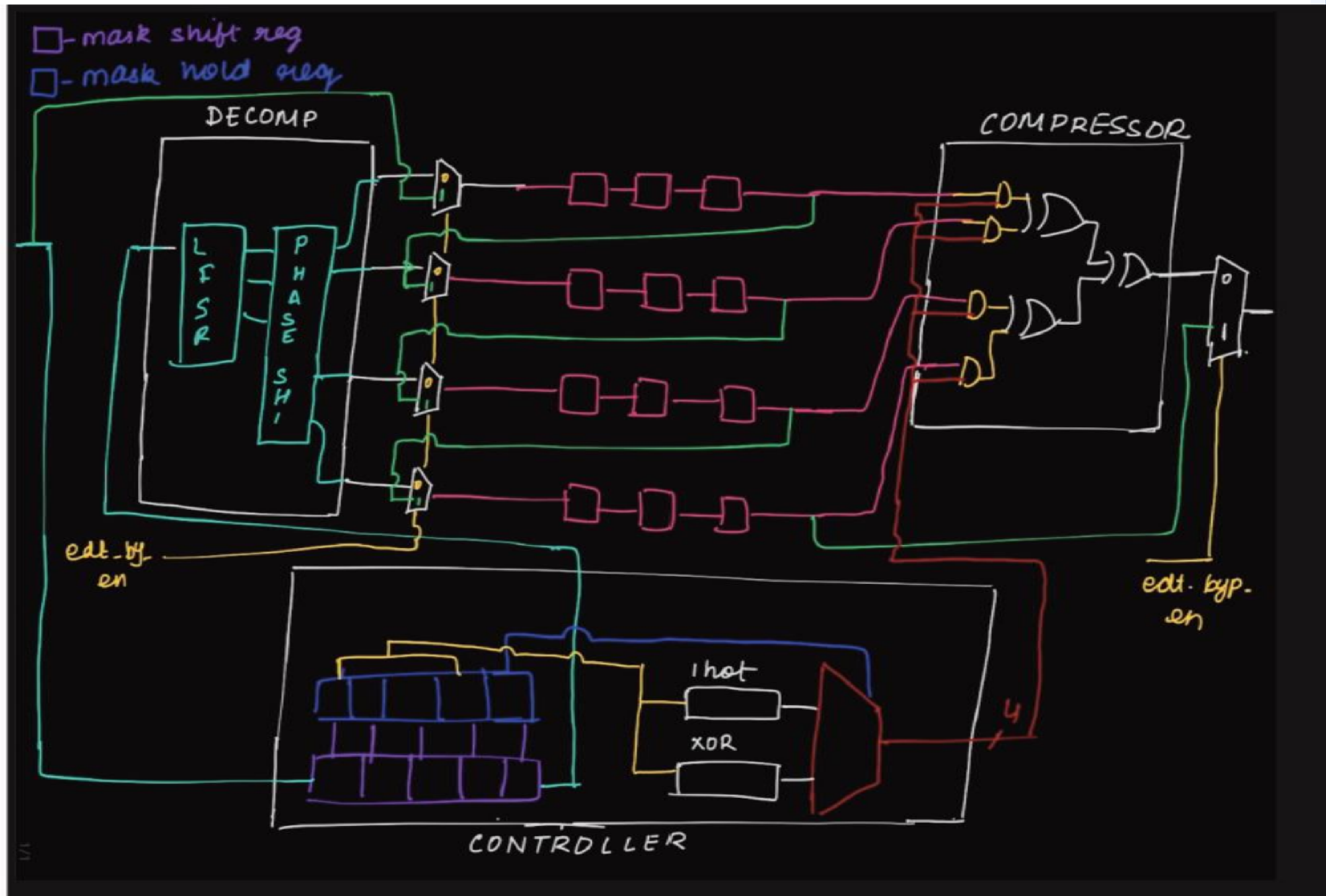


EDT OUTPUTS



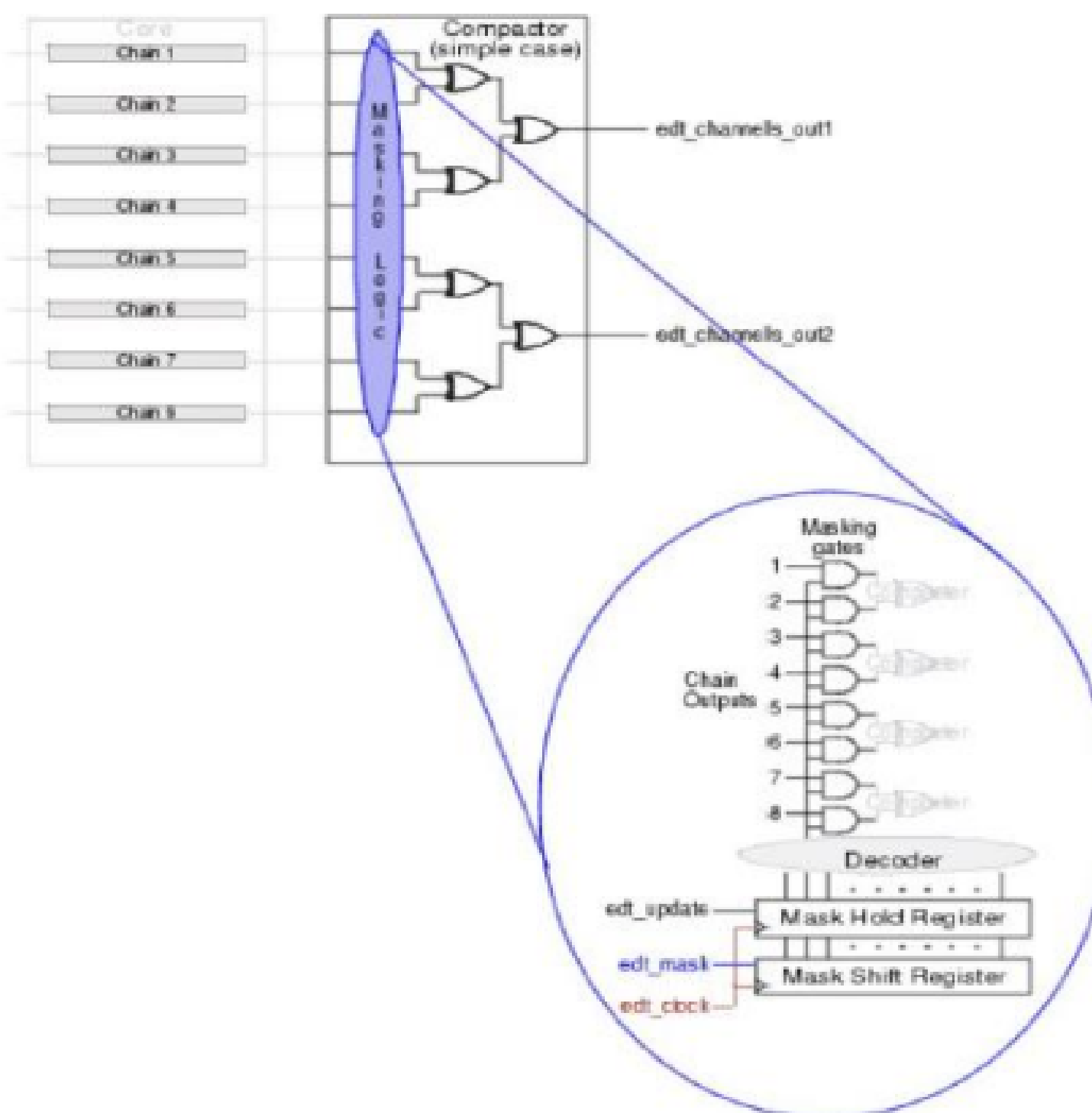
edt.dofile Explanation

Command : set_edt_options

-compactor_type xpress

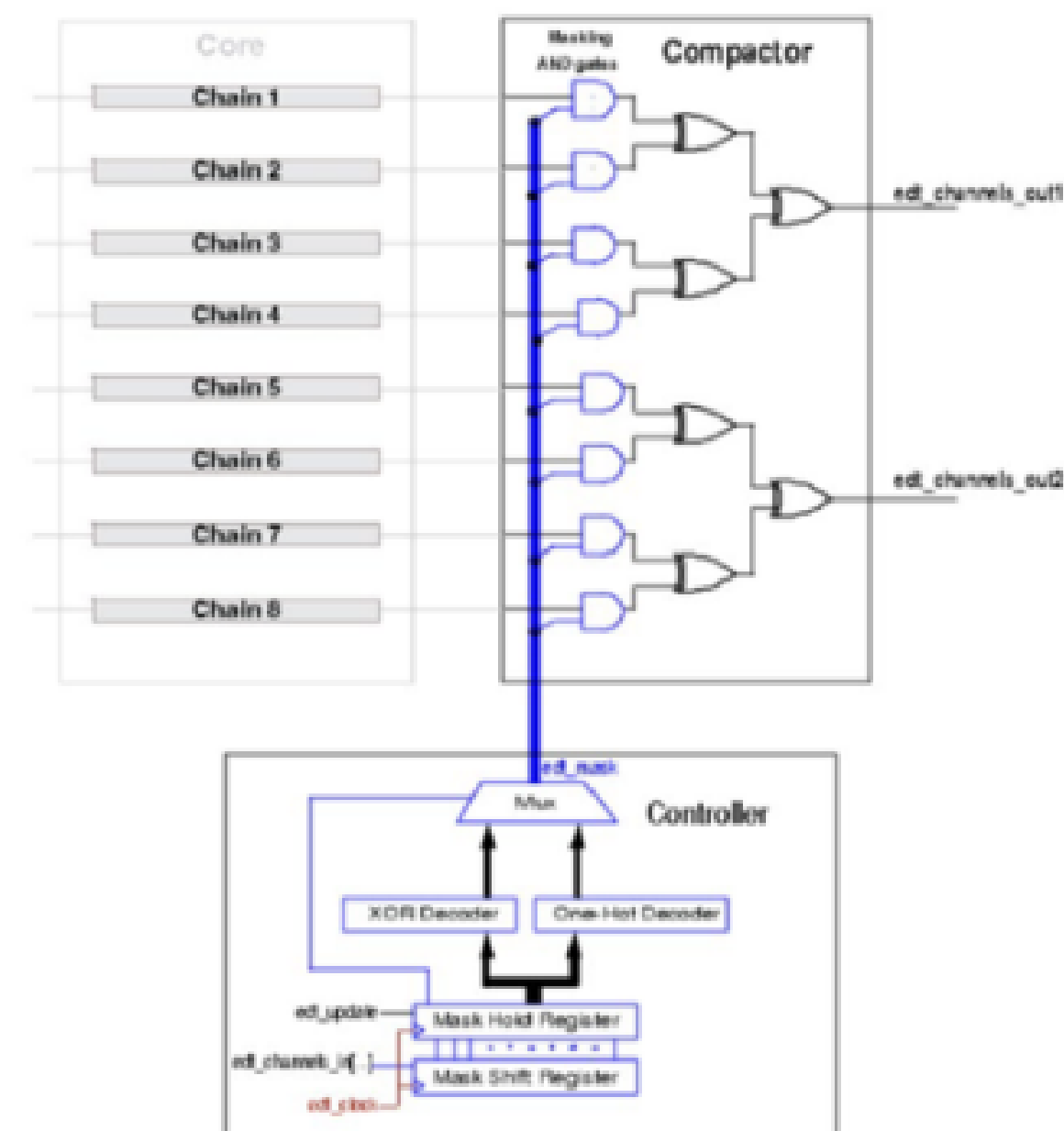
Basic compactor : Allows one scan chain or all scan chains based on the mode bit.

BASIC COMPACTOR :



Xpress Compactor : It chooses between 1-hot decoder and XOR decoder

XPRESS COMPACTOR :



edt.dofile Explanation (Contd...)

-lockup_cells on/off

on - Includes lockup cells in the EDT logic

off - Disables lockup cells in the EDT logic.

set_edt_pins output_channel -change_edge_at_compactor_output trailing_edge_of_edt_clock

-change edge at compactor output

trailing_edge_of_edt_clock : Specifies that compactor outputs need to change on the TE edge of the EDT clock

leading_edge_of_edt_clock : Specifies that all compactor outputs need to change on the LE edge of the EDT clock.

Any — Specifies that data can change at either edge.

for which ffs in the LFSR logic,
the external i/p is given

edt.dofile

set_decompressor_connections -channel 1 -taps 1 3 7 5

RTL

```
lfsm_vec[0] <= lfsm_vec[1];  
lfsm_vec[1] <= lfsm_vec[2] ^ edt_channels_in;  
lfsm_vec[2] <= lfsm_vec[3];  
lfsm_vec[3] <= lfsm_vec[4] ^ lfsm_vec[3] ^ edt_channels_in;  
lfsm_vec[4] <= lfsm_vec[5] ^ lfsm_vec[2] ^ lfsm_vec[3];  
lfsm_vec[5] <= lfsm_vec[6] ^ lfsm_vec[2] ^ edt_channels_in;  
lfsm_vec[6] <= lfsm_vec[7] ^ lfsm_vec[1];  
lfsm_vec[7] <= lfsm_vec[0] ^ edt_channels_in;
```

```
set_mask_register -input_channel_mask_register_sizes
```

which channel is
controlling the
mask shift reg and
mask hold reg

no. of ffs
in mask shift reg
and mask hold
reg

controlling the mask hold reg

edt.dofile

```
set_mask_decoder_connection -mode_bit
```

RTL

```
assign control_bit = masks_hold_reg_0[4];
```

which
mask hold
reg's O/P

is used as a
control bit to sel blw
1-hot and x-masking
decoder

Note:

In RTL,

mask_hold_reg[0]	→	1 st flop
mask_hold_reg[1]	→	2 nd ff
mask_hold_reg[2]	→	3 rd ff
mask_hold_reg[3]	→	4 th ff
mask_hold_reg[4]	→	5 th ff

In edt.dofile

0 → which channel is controlling the respective mask hold reg

0 → which mask hold reg is going as a i/p to 1-hot decoder

```
set_mask_decoder_connection -lhot_decoder 1
set_mask_decoder_connection -xor_decoder chain1
set_mask_decoder_connection -xor_decoder chain2
set_mask_decoder_connection -xor_decoder chain3
set_mask_decoder_connection -xor_decoder chain4
```

```
assign xor_encoded_masks = {masks_hold_reg_0[0], masks_hold_reg_0[1], masks_hold_reg_0[2],
                             masks_hold_reg_0[3]};
```

```
assign onehot_encoded_masks_0 = {masks_hold_reg_0[3], masks_hold_reg_0[2], masks_hold_reg_0[1]};
```

```
DmaWr_edt_xor_decoder xor_decoder (.encoded_masks(xor_encoded_masks),
                                     .decoded_masks_0(xor_decoded_masks_0));
```

```
DmaWr_edt_onehot_decoder_3_to_4 decoder1 (.encoded_masks(onehot_encoded_masks_0),
                                           .decoded_masks(onehot_decoded_masks_0));
```

0 → which mask hold reg's O/P is going as a i/p to xor decoder

NOTE

INSERTING PIPELINE FLOPS IN THE COMPACTOR

set_edt_options → Command

Switch -> -Pipeline_logic_levels_in_compactor {OFF | integer}

- OFF — Disables pipeline stages in the EDT logic.
- integer — Includes pipeline stages in the EDT logic compactor(s)

Example : set_edt_options -input_channels 1 -output_channels 1 -location internal -Pipeline_logic_levels_in_compactor 1
(command to be given in script file. It will insert 1 pipeline stage in compactor)